RACE Regular Analysis Through Continuous Exercise

Complex Numbers : RACE-01

Subjective Type Question

1. Find the value of the sum $\sum_{n=1}^{5}(i^n + i^{n+2})$.

MCN231

2. Find the value of $13\left(1+\frac{2i}{3}+\left(\frac{2i}{3}\right)^2+\left(\frac{2i}{3}\right)^3+...\infty\right)$.

MCN232

3. Express the set $A = \{i^n + i^{-n} : n \in \mathbb{Z}\}$ in roster form.

MCN233

4. Find the absolute value of $\frac{i^{592}+i^{590}+i^{588}+i^{586}+i^{584}}{i^{582}+i^{580}+i^{578}+i^{576}+i^{574}}$.

MCN234

5. Given that $x, y \in \mathbb{R}$. Solve the following equations and find (x, y).

(a) $(x + 2y) + i\,(2x - 3y) = 5 - 4i$ (b) $(x + iy) + (7 - 5i) = 9 + 4i$

(c) $x^2 - y^2 - i\,(2x + y) = 2i$ (d) $\frac{x-3}{3+i}+\frac{y-3}{3-i}=i$

MCN235

6. If $x+iy=\frac{1}{1+\cos\theta+i\sin\theta}$, then find the value of $\frac{2x-1+3y}{4x-2+5y}$.

MCN236

7. If $x+iy=\left(\frac{a+ib}{a-ib}\right)^2-\left(\frac{a-ib}{a+ib}\right)^2$, then find the value of $2x + y$; given that $x, y, a, b \in \mathbb{R}$.

MCN237

8. If $a + ib = \frac{c+i}{c-i}$, where $a,b,c \in \mathbb{R}$, then find the value of $\frac{2ac\left(a^2+b^2\right)}{b\left(c^2-1\right)}$.

MCN238

9. Simplify the following complex numbers and express the result in the form of $a + bi$ where $a, b \in \mathbb{R}$

(a) $\left(\frac{1+2i}{2+i}\right)^2$ (b) $-i(9 + 6i)(2 - i)^{-1}$ (c) $\left(\frac{4i^3-i}{2i+1}\right)^2$ (d) $\frac{(2+i)^2}{2-i}-\frac{(2-i)^2}{2+i}$

MCN239

For More Join: @IITJEE_Advanced

10. Determine least positive integral value of n for which $\left(\frac{1+i}{1-i}\right)^n = 1$.

MCN240

11. Find the square root(s) of following complex numbers :

(a) $9 + 40i$ (b) $-11 - 60i$ (c) $50i$ (d) $-15 - 8i$

MCN241

12. Solve the following equations for x and express the solution in the form $a + ib$, $a, b \in \mathbb{R}$.

(a) $ix^2 - 3x - 2i = 0$ (b) $2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$

MCN242

13. If $(x+iy)^{\frac{1}{3}} = a + bi$, then find $\frac{\frac{x}{a}+\frac{y}{b}}{a^2-b^2}$; given that $x, y, a, b \in \mathbb{R}$.

MCN243

14. If $x = 1 + 2i$, then find the value of $13x - x^3 - 7x^2 - 16$.

MCN244

15. If $f(x) = x^4 + 9x^3 + 35x^2 - x + 4$ and $g(x) = x^4 - x^3 + x^2 + 3x - 5$, then find the value of $g(2 + 3i) - f(-5 + 4i)$

MCN245

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ANSWER KEY

1. 0 **2.** $9 + 6i$ **3.** $0, 2, -2$ **4.** 1

5. (a) (1,2); (b) (2,9); (c) $(-2,2)$ or $\left(-\frac{2}{3}, -\frac{2}{3}\right)$; (d) (−2,8) **6.** $\frac{3}{5}$

7. $\frac{8ab\left(a^2-b^2\right)}{\left(a^2+b^2\right)^2}$ **8.** 1 **9.** (a) $\frac{7}{25}+\frac{24}{25}i$; (b) $\frac{21}{5}-\frac{12}{5}i$; (c) $3 + 4i$; (d) $\frac{22}{5}i$;

10. $n = 4$ **11.** (a) $\pm(5 + 4i)$; (b) $\pm(5 - 6i)$; (c) $\pm 5(1 + i)$; (d) $\pm(1 + 4i)$

12. (a) $-i, -2i$ (b) $\frac{3-5i}{2}$ or $-\frac{1+i}{2}$ **13.** 4 **14.** 29 **15.** $83 - 108i$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Complex Numbers : RACE-02

Subjective Type Question

1. If $z_1 = 1 + 2i$ and $z_2 = 3 + 5i$, and then find $\text{Re}\left(\frac{\bar{z}_2 z_1}{z_2}\right)$.

MCN246

2. If $z = \left(\frac{4i^3 - i}{2i + 1}\right)^2 + \frac{1}{1 - \cos\theta + 2i\sin\theta}; \theta \in \mathbb{R}$, then find the value of $2\left(\frac{z + \bar{z}}{2}\right) + 3\left(\frac{z - \bar{z}}{2i}\right)$.

MCN247

3. If $z = \frac{2 + i}{x + 3 + iy}; x, y \in \mathbb{R}$ then find $\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2}$ in terms of x, y.

MCN248

4. If the complex number $\frac{\sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right) + i\tan(x)}{1 + 2i\sin\left(\frac{x}{2}\right)}$ is equal to its conjugate, then find the set of all possible values of x.

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MCN249

5. If $z^4 + 4 = (z - \alpha)(z - \bar{\alpha})(z - \beta)(z - \bar{\beta})$, $\text{Re}(\alpha) > 0, \text{Re}(\beta) < 0$, $\text{Im}(\alpha) > 0, \text{Im}(\beta) < 0$, then find α and β.

MCN250

6. Prove that $z_1\left(\overline{z_2 + z_3}\right) + z_2\left(\overline{z_3 + z_1}\right) + z_3\left(\overline{z_1 + z_2}\right) = 2\text{Re}\left(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1\right) = 2\text{Re}\left(\bar{z}_1 z_2 + \bar{z}_2 z_3 + \bar{z}_3 z_1\right)$

MCN251

7. If the equation, $z^2 - (3 + i)z + m + 2i = 0; m \in \mathbb{R}$ has a real root, then find m.

MCN252

8. If a, b, c, a_1, b_1, c_1, are non-zero complex numbers satisfying $\frac{a}{a_1} + \frac{b}{b_1} + \frac{c}{c_1} = 1 + i$ and $\frac{a_1}{a} + \frac{b_1}{b} + \frac{c_1}{c} = 0$, then prove that $\frac{a^2}{a_1^2} + \frac{b^2}{b_1^2} + \frac{c^2}{c_1^2} = 2i$.

MCN253

9. Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers. Find the condition that the equation has one purely imaginary root.

MCN254

10. Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers. Find the condition that the equation has two purely imaginary roots.

MCN255

For More Join: @IITJEE_Advanced

11. Given that the equation $z^2 + uz + v = 0$ where u,v are non-zero complex numbers has a real root. Then prove that $(u+\bar{u})(u-\bar{u})(v-\bar{v}) = 2(v-\bar{v})^2 + (u-\bar{u})^2(v+\bar{v})$.

MCN256

12. Find the number of real $z \in \{x+iy : x, y \in \mathbb{R}, xy = 0\}$ satisfying $\bar{z}^3 - i \cdot \bar{z} - 1 = 0$

MCN257

13. If a, b, c are real numbers and z is a complex number such that $a^2 + b^2 + c^2 = 1$ and $b + ic = (1 + a)z$, then prove that $\frac{1 - i\bar{z}}{1 + i\bar{z}} = \frac{a - ib}{1 + c}$.

MCN258

14. Let a, b, c are real numbers in the polynomial $P(z) = 2z^4 + az^3 + bz^2 + cz + 3$. If two roots of the equation $P(z) = 0$ are 2 and i, then find the value of 'a'.

MCN259

15. Solve the equation $z^2 - (3 - 2i)z = (5i - 5)$ and find z ($\in \mathbb{C}$).

MCN260

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ANSWER KEY

1. $\frac{22}{17}$ **2.** $\frac{92 + 54\cos\theta - 6\cot\frac{\theta}{2}}{5 + 3\cos\theta}$ **3.** $\left(\frac{2(x+3)+y}{(x+3)^2+y^2}, \frac{(x-2y+3)i}{(x+3)^2+y^2}\right)$

4. $2n\pi, n\pi + \frac{\pi}{4}$; $n \in \mathbb{Z}$

5. $z^4 + 4 = (z + 1 + i)(z + 1 - i)(z - 1 + i)(z - 1 - i)$; $\alpha = 1 + i$ and $\beta = -1 - i$

7. 2 **9.** $(b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b) + (c\bar{a} - a\bar{c})^2 = 0$ **10.** $\frac{a}{\bar{a}} = -\frac{b}{\bar{b}} = \frac{c}{\bar{c}}$

12. 0 **14.** $-\frac{11}{2}$ **15.** $z = (2 + i)$ or $(1 - 3i)$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Complex Numbers : RACE-03

Subjective Type Question

1. If $(1-i)(1-2i)\ldots(1-ni) = x - iy$; $x, y \in \mathbb{R}$ then prove that $x^2 + y^2 = \prod_{r=1}^{n}\left(r^2+1\right)$

MCN261

2. If $x+iy = \sqrt{\frac{a-ib}{c-id}}$, then prove that $\left(x^2+y^2\right)^2 = \frac{a^2+b^2}{c^2+d^2}$; $a, b, c, d, x, y \in \mathbb{R}$.

MCN262

3. If $a_1, a_2, \ldots, a_n$ are the real roots of the polynomial $x^n + p_1x^{n-1} + p_2x^{n-2} + \ldots + p_{n-1}x + p_n$ ($p_1, p_2, \ldots, p_n$ are real), then prove that

$\left(1+a_1^2\right)\left(1+a_2^2\right)\ldots\left(1+a_n^2\right) = \left(1-p_2+p_4-\ldots\right)^2 + \left(p_1-p_3+p_5-\ldots\right)^2$.

MCN263

4. If $z_1, z_2 \in \mathbb{C}$, $z_1^2 + z_2^2 \in \mathbb{R}$, $z_1\left(z_1^2 - 3z_2^2\right) = 2$ and $z_2\left(3z_1^2 - z_2^2\right) = 11$, then find $z_1^2 + z_2^2$.

MCN264

5. If 'x' and 'a' are real numbers, then prove that

$\left(x^2+a^2\right)^7 = \left(x^7 - 21x^5a^2 + 35x^3a^4 - 7xa^6\right)^2 + \left(7x^6a - 35x^4a^3 + 21x^2a^5 - a^7\right)^2$.

MCN265

6. Using complex numbers, prove that

$(1^2 + 2^2)(3^2 + 4^2)(5^2 + 7^2) = 95^2 + 15^2 = 45^2 + 85^2 = 41^2 + 87^2 = 69^2 + 67^2$.

MCN266

7. If z_1, z_2, z_3 be three complex numbers such that $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$, then find $|9z_1z_2 + 4z_1z_3 + z_2z_3|$.

MCN267

8. For any two complex numbers z_1, z_2 and any two real numbers a and b, prove that

$|az_1 - bz_2|^2 + |bz_1 + az_2|^2 = (a^2 + b^2)(|z_1|^2 + |z_2|^2)$.

MCN268

9. Show using the properties of modulus that $\left|2\bar{z}+5\right|\left(\sqrt{2}-i\right)\left| = \sqrt{3}\right|2z+5|$, where $z \in \mathbb{C}$.

MCN269

10. If $\frac{a}{|z_1 - z_2|} = \frac{b}{|z_2 - z_3|} = \frac{c}{|z_3 - z_1|}$; $a, b, c \in R$, and $z_1, z_2, z_3 \in \mathbb{C}$, then prove that

$$\frac{a^2}{z_1 - z_2} + \frac{b^2}{z_2 - z_3} + \frac{c^2}{z_3 - z_1} = 0.$$

MCN270

For More Join: @IITJEE_Advanced

11. Find the modulus, argument and the principal argument of $\frac{2+i}{4i+(1+i)^2}$

MCN271

12. If $z \in \mathbb{C}, |z| = 1, \arg(z) = \theta$, then find $\text{Re}\left(\frac{z(1-\bar{z})}{\bar{z}(1+z)}\right)$ in terms of θ.

MCN272

13. Let z, w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and amp $(zw) = \pi$, then find amp(z). (Here amp(z) denotes the principle value of argument.)

MCN273

14. Let amp(z) denote the principle value of argument. If z and w are non zero complex numbers such that $|zw| = 1$ and amp(z) – amp$(w) = \frac{\pi}{2'}$ then find $\bar{z}w$.

MCN274

15. Let amp(z) denote the principle value of argument. If z and w are two non-zero complex numbers such that $|z| = |w|$ and amp z + amp $w = \pi$, then prove that $z = -\bar{\omega}$

MCN275

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ANSWER KEY

4. 5 **7.** 6 **11.** Modulus = $\frac{\sqrt{5}}{6}$, Arg = $2k\pi - \tan^{-1}2 (k \in \mathbb{Z})$. Principal Arg = $-\tan^{-1}2$

12. $-2\sin^2\frac{\theta}{2}$ **13.** $\frac{3\pi}{4}$ **14.** $-i$

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Complex Numbers : RACE-04

Subjective Type Question

1. Express the following complex numbers in polar form and exponential form :

(i) $-2 + 2i$ (ii) $-1-\sqrt{3}i$

(iii) $\frac{(1+7i)}{(2-i)^2}$ (iv) $(1 - \cos\theta + i\sin\theta);\ \theta \in (0,\pi)$

MCN276

2. Express the following complex numbers in the polar form :

(i) $\frac{1+i}{1-i}$ (ii) $\frac{2+6\sqrt{3}i}{5+\sqrt{3}i}$

MCN277

3. Prove that $\frac{(\cos\theta - i\sin\theta)^4}{(\sin\theta + i\cos\theta)^5} = \sin\theta - i\cos\theta\ \forall\theta \in \mathbb{R}$.

MCN278

4. Find the value of $\frac{4(i\cos 11^\circ - \sin 11^\circ)}{0.4(i\cos 34^\circ + \sin 34^\circ)}$ using Euler's formula.

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MCN279

5. Let amp(z) denote the principle value of argument. If z is a non-real complex number satisfying $|z| = 1$, then prove that $z = \frac{1 + i\tan\left(\frac{amp(z)}{2}\right)}{1 - i\tan\left(\frac{amp(z)}{2}\right)}$.

MCN280

6. If z is a complex number such that $|z| = r, \arg(z) = \theta$, then find $\left|e^{iz}\right|$ in terms of r and θ.

MCN281

7. If α, β are the roots of the equation $x^2 - 2x + 4 = 0$, then prove that $\alpha^n + \beta^n = 2^{n+1}\cos\frac{n\pi}{3}$ for all $n \in \mathbb{N}$.

MCN282

8. If $n \in \mathbb{N}$, then prove that $\left(\frac{1+\sin\phi+i\cos\phi}{1+\sin\phi-i\cos\phi}\right)^n = \cos\left(\frac{n\pi}{2} - n\phi\right) + i\sin\left(\frac{n\pi}{2} - n\phi\right) \forall\phi \in \mathbb{R}$.

MCN283

9. Let z be a complex number with non-zero imaginary part and $|z| = 1$. Show that z can be represented as $\frac{c+i}{c-i}$ where c is real.

MCN284

For More Join: @IITJEE_Advanced

10. If $z = \cos\frac{\pi}{8} + i\sin\frac{\pi}{8}$ and $A = \{z^n : n \in N\}$, then find $n(A)$.

MCN285

11. If 'a' is a complex number such that $|a| = 1$, find the values of a, so that equation $az^2 + z + 1 = 0$ has one purely imaginary root.

MCN286

12. If z is a complex number such that $z + \frac{1}{z} = 2\cos3°$, and $\omega = \left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^{15}$, then find absolute value of $z^{2000} + \frac{1}{z^{2000}} + \text{Im}(\omega)$.

MCN287

13. If $\cos\alpha + \cos\beta + \cos\gamma = 0$ and $\sin\alpha + \sin\beta + \sin\gamma = 0$, then prove the following :

(a) $\sum\sin(\alpha+\beta) = 0 = \sum\cos(\alpha+\beta)$

(b) $\sum\sin^2\alpha = \frac{3}{2} = \sum\cos^2\alpha$

(c) $\cos(\alpha-\beta) + \cos(\beta-\gamma) + \cos(\gamma-\alpha) = \frac{-3}{2}$

MCN288

14. Prove that $e^{2mi\cot^{-1}p}\left(\frac{pi+1}{pi-1}\right)^m = \frac{(1-i)^{20}\left(\sqrt{3}+i\right)^{10}}{\left(-1-i\sqrt{3}\right)^{20}}$; where $p > 0$ and $m \in \mathbb{Z}$.

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MCN289

15. If $z_r = \cos\frac{2r\pi}{5} + i\sin\frac{2r\pi}{5}$, $r = 0, 1, 3, 4, ...$, then find $z_1z_2z_3z_4z_5$.

MCN290

ANSWER KEY

1. (i) $2\sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right); 2\sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$; (ii) $2\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right); 2e^{i\left(\frac{4\pi}{3}\right)}$;

(iii) $\sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right); \sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$; (iv) $2\sin\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\pi}{2}-\frac{\theta}{2}\right) + i\sin\left(\frac{\pi}{2}-\frac{\theta}{2}\right)\right); 2\sin\left(\frac{\theta}{2}\right)e^{i\left(\frac{\pi}{2}-\frac{\theta}{2}\right)}$

2. (i) $z = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2}$ (ii) $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

4. $\frac{10}{\sqrt{2}}(1+i)$ **6.** $e^{-r\sin\theta}$ **10.** 16

11. $a = \cos\theta + i\sin\theta$, where $\theta = \cos^{-1}\left(\frac{\sqrt{5}-1}{2}\right)$ **12.** 2 **15.** 1

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Complex Numbers : RACE-05

Subjective Type Question

1. Locate the points representing the complex number z on the Argand plane satisfying the following equations :

(a) $|z+1-2i|=\sqrt{7}$ (b) $|z-1|^2+|z+1|^2=4$

(c) $\left|\frac{z-3}{z+3}\right|=3$ (d) $|z-3|=|z-5+6i|$

MCN291

2. Locate the points representing the complex number z on the Argand plane satisfying the following equations :

(a) $|z-11-2i|<5$ (b) $|z-12i-10|>10$

(c) $10<|z-12+10i|<20$ (d) $5\le|z-2-3i|\le7$

MCN292

3. Find center and radius of circle represented by $z\bar{z}+(2-3i)z+(2+3i)\bar{z}+4=0$

MCN293

4. If z_1, z_2, z_3, z_4 are the affixes of four points in the Argand plane and z is the affix of a point such that $|z-z_1|=|z-z_2|=|z-z_3|=|z-z_4|$, then prove that z_1, z_2, z_3, z_4 are concyclic,

MCN294

5. Locate the points representing the complex number z on the Argand plane satisfying the following equations :

(a) $|z-1|+|z+1|=2$ (b) $|z-i|+|z+1|=2$

(c) $|z-12i|+|z+5|<20$ (d) $|z-12|+|z+5i|=10$

MCN295

6. Locate the points representing the complex number z on the Argand plane satisfying the following equations :

(a) $\big||z-i|-|z+2i|\big|=3$ (b) $\big||z-4i|-|z+3|\big|=2$

(c) $|z-12i|-|z+5|=20$ (d) $|z-8i|-|z+6|=5$

MCN296

7. Find range of k, if the points z satisfying $|z+i|-|z-i|=k$ lie on a hyperbola.

MCN297

8. If locus of z satisfying $|z-i|+|z+2i|=k, k>0$, is an ellipse, then find range of k^2.

MCN298

9. Prove that the locus of $P(z)$ satisfying $\left|\frac{z-z_1}{z-z_2}\right|=k$ $(k\neq1, >0)$ is a circle whose center divides $A(z_1)$ and $B(z_2)$ in ratio $k^2:1$ externally.

MCN299

10. Let z_1, z_2 are complex numbers satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$. Find the minimum value of $|z_1 - z_2|$.

MCN300

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11. If the complex number z satisfies $|z| \geq 3$, then find the least value of $\left|z+\frac{1}{z}\right|$.

MCN301

12. If z is any complex number satisfying $|z-3-2i| \leq 2$, then find the minimum value of $|2z-6+5i|$.

MCN302

13. Find the number of solutions of the equations
$|z-(4+8i)|=\sqrt{10}$ and $|z-(3+5i)|+|z-(5+11i)|=4\sqrt{5}$.

MCN303

14. Let $|z| = 2$ and $w=\frac{z+1}{z-1}$ where $z, w \in C$. If M and m respectively be the greatest and least modulus of w, then find the value of $(2010m + M)$.

MCN304

15. If r_1 and r_2 are the maximum and minimum distances of a point on the curve $10(z\bar{z})-3i\{z^2-(\bar{z})^2\}-6=0$ from origin, then find $r_1 + r_2$.

MCN305

ANSWER KEY

1. (a) Circle of radius $\sqrt{7}$ with centre $(-1, 2)$
(b) Unit circle with centre at origin
(c) Circle with centre $\left(-\frac{15}{4},0\right)$ and radius $\left(\frac{9}{4}\right)$
(d) Straight line $x - 3y = 13$

2. (a) Inside a circle with center (11, 2) and radius 5.
(b) Outside a circle with center (10, 12) and radius 10.
(c) Between two concentric circles with center (12, –10) and radii 10, 20.
(d) On or between two concentric circles with center (2, 3) and radii 5, 7.

3. Center = (–2, –3), Radius = 3

5. (a) Line segment AB, where $A(1, 0)$ and $B(-1, 0)$
(b) Ellipse with eccentricity $\frac{1}{\sqrt{2}}$ and foci $A(0, 1)$, $B(-1, 0)$
(c) Inside an ellipse with eccentricity $\frac{13}{20}$ and foci $A(0, 12)$, $B(-5, 0)$
(d) No such point exists

6. (a) Points on line AB, where $A(1, 0)$ and $B(-1, 0)$ excluding those between A and B.
(b) Hyperbola with eccentricity $\frac{5}{2}$ and foci $A(0, 4)$,$B(-3, 0)$
(c) No such point exists
(d) Branch of Hyperbola with eccentricity 2 and foci $A(0, 8)$, $B(-6, 0)$ closer to B.

7.	$-2 < k < 2, k \neq 0$	**8.**	$9, \infty$	**10.**	2	**11.**	$\frac{8}{3}$
12.	5	**13.**	2	**14.**	673	**15.**	3

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RACE
Regular Analysis Through Continuous Exercise

Complex Numbers : RACE-06

Single Correct Question

1. If z is a complex number, then locus of z satisfying $z^2 + \bar{z}^2 = 2$ is
(A) a circle (B) a straight line (C) a hyperbola (D) an ellipse

MCN306

Subjective Type Question

2. Find the number of solutions of $\text{Re}(z^2) = 0$ and $|z| = a\sqrt{2}$, where z is a complex number and $a > 0$.

MCN307

3. Let z be a complex number such that $z = (1-t) + i\sqrt{t^2+t+2}$ where t is a real parameter. Then prove that the locus of z on Argand Plane is a hyperbola.

MCN308

4. If $z \neq 1$ and $\frac{z^2}{z-1}$ is real, then prove that the point represented by the complex number z lies either on the real axis or on a circle passing through the origin

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MCN309

5. Prove that $\left\{\text{Re}\left(\frac{2iz}{1-z^2}\right) : z \text{ is a complex number, } |z| = 1, z \neq \pm 1\right\} = (-\infty, -1] \cup [1, \infty)$.

MCN310

6. Find all complex numbers z satisfying the equation $2|z|^2 + z^2 - 5 + i\sqrt{3} = 0$.

MCN311

7. Let $z = x + iy$ be a complex number where x and y are integers. Then find the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$.

MCN312

8. Let z be a complex number such that $z \in \mathbb{C} \setminus \mathbb{R}$ and $\frac{1+z+z^2}{1-z+z^2} \in R$, then prove that $|z| = 1$.

MCN313

9. Find the minimum value of the expression $E = |z|^2 + |z-3|^2 + |z-6i|^2$
(where $z = x + iy, x, y \in R$)

MCN314

10. If z_1 and z_2 are two given complex numbers, then prove that the locus of z satisfying $|z-z_1|^2 + |z-z_2|^2 = k$, is a circle having centre at $\frac{z_1+z_2}{2}$ and radius = $\frac{1}{2}\sqrt{2k - |z_1 - z_2|^2}$.

MCN315

For More Join: @IITJEE_Advanced

11. If A, B, C & D are four points representing the complex numbers z_1, z_2, z_3 & z_4 then prove that $AB \parallel CD$ if $\frac{z_4 - z_3}{z_2 - z_1}$ is purely real.

MCN316

12. If A, B, C & D are four points representing the complex numbers z_1, z_2, z_3 & z_4 then prove that $AB \perp CD$ if $\frac{z_4 - z_3}{z_2 - z_1}$ is purely imaginary.

MCN317

13. If $\bar{\alpha} z + \alpha \bar{z} + r = 0$, where r is real and α is non zero complex constant represents the equation of perpendicular bisector of the line segment joining $A(z_1)$ and $B(z_2)$, then prove that $\frac{\bar{\alpha}}{\bar{z}_1 - \bar{z}_2} = \frac{\alpha}{z_1 - z_2} = \frac{-r}{|z_1|^2 - |z_2|^2}$.

MCN318

14. Prove that the perpendicular distance of the point z_c from the line $z\bar{a} + \bar{z}a + b = 0$, $(z, a \in \mathbb{C}, b \in \mathbb{R})$ is $\frac{|z_c \bar{a} + \bar{z}_c a + b|}{2|a|}$

MCN319

15. If ω denotes the orthocentre of ΔABC where $A(z_1), B(z_2), C(z_3)$, then prove that

$$\frac{\omega - z_1}{\bar{\omega} - \bar{z}_1} + \frac{\omega - z_2}{\bar{\omega} - \bar{z}_2} + \frac{\omega - z_3}{\bar{\omega} - \bar{z}_3} + \frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1} + \frac{z_2 - z_3}{\bar{z}_2 - \bar{z}_3} + \frac{z_3 - z_1}{\bar{z}_3 - \bar{z}_1} = 0$$

Rankers Academy JEE

MCN320

ANSWER KEY

1.	C	**2.**	4	**6.**	$\frac{1}{\sqrt{6}} - \frac{3}{\sqrt{2}}i, -\frac{1}{\sqrt{6}} + \frac{3}{\sqrt{2}}i, \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}i, -\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}i$
7.	48	**9.**	30		

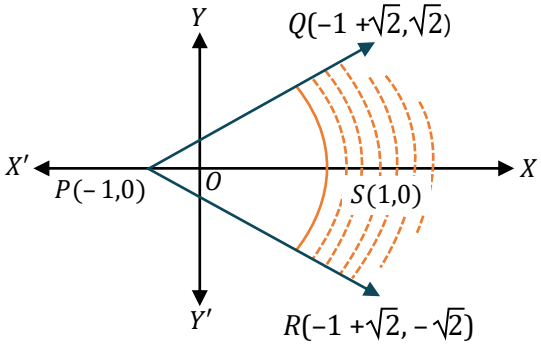
For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Complex Numbers : RACE-07

Single Correct Question

1. The shaded region, where $P \equiv (-1, 0)$, $Q \equiv (-1+\sqrt{2}, \sqrt{2})$, $R \equiv (-1+\sqrt{2}, -\sqrt{2})$ and $S \equiv (1, 0)$ is represented by : (Here amp(z) denotes the principle value of argument.)



(A) $|z + 1| > 2, |\text{amp}(z + 1)| < \frac{\pi}{4}$ (B) $|z + 1| < 2, |\text{amp}(z + 1)| < \frac{\pi}{2}$

(C) $|z - 1| > 2, |\text{amp}(z + 1)| > \frac{\pi}{4}$ (D) $|z - 1| < 2, |\text{amp}(z + 1)| > \frac{\pi}{2}$

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MCN321

Subjective Type Question

2. Let amp(z) denote the principle value of argument. Sketch the locus of z satisfying the following equations:

(i) $\text{amp}(z) = -\frac{5\pi}{6}$ (ii) $\text{amp}(z+i) = -\frac{3\pi}{4}$

(iii) amp$(z + 3 + 4i) = \frac{\pi}{2}$ (iv) amp$(z - 7 - 4i) = \pi$ (v) $amp(z+7) = \frac{3\pi}{4}$

MCN322

3. Let amp(z) denote the principle value of argument. If $|z - 25i| \le 15$, then find $|\max(\text{amp}z) - \min(\text{amp}z)|$.

MCN323

4. Let arg(z) denote the principle value of argument. Prove that points z satisfying the equation $|z-2-i| = |z|.\left|\sin\left(\frac{\pi}{4} - \arg(z)\right)\right|$ lie on a parabola.

MCN324

5. Find the area of the region of the argand plane described by complex numbers z satisfying $\frac{\pi}{6} < \text{amp}(z) < \frac{2\pi}{3}$ & $3 < |z| < 5$. Here amp(z) denotes the principle value of argument.

MCN325

For More Join: @IITJEE_Advanced

6. For any two complex numbers z_1, z_2 prove the following (Here arg(z) denotes the principle value of argument.)

(i) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2|z_1||z_2| \cos(\theta_1 - \theta_2)$ (θ_1, θ_2 are arguments of z_1, z_2)

(ii) $|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2|z_1||z_2| \cos(\theta_1 - \theta_2)$ (θ_1, θ_2 are arguments of z_1, z_2)

(iii) $|z_1 + z_2| = |z_1 - z_2| \Leftrightarrow \arg(z_1) - \arg(z_2) = \pm\frac{\pi}{2} \Leftrightarrow \frac{z_1}{z_2}$ is purely imaginary ($z_2 \neq 0$)

(iv) $|z_1 + z_2| = |z_1| + |z_2| \Leftrightarrow \arg(z_1) = \arg(z_2) \Leftrightarrow \frac{z_1}{z_2}$ is purely imaginary ($z_2 \neq 0$)

(v) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \Leftrightarrow \frac{z_1}{z_2}$ is purely imaginary ($z_2 \neq 0$)

MCN326

7. Among the complex numbers z satisfying the condition $\left|z+3-\sqrt{3}i\right| = \sqrt{3}$, find the number having the least positive principle argument.

MCN327

8. A complex number $z = 3 + 4i$ is rotated about another fixed complex number $z_1 = 1 + 2i$ in anticlockwise direction by 45° angle. Find the complex number represented by new position of z in argand plane.

MCN328

9. Let A, B, C are three points in argand plane (in anticlockwise sense) representing the complex numbers z_1, z_2, z_3. Find conditions on ℓ, m, n, if $\frac{\ell z_1 + m z_2 + n z_3}{\ell + m + n}$ lies outside ΔABC, but inside $\angle ABC$.

Rankers Academy JEE

MCN329

10. If $A(z_1), B(z_2), C(z_3)$ are the vertices of an equilateral ΔABC, taken anticlockwise then find amp $\left(\frac{z_2 + z_3 - 2z_1}{z_3 - z_2}\right)$

MCN330

11. Find the center of the arc represented by locus of points z satisfying $amp\left(\frac{z-3i}{z-2i+4}\right) = \frac{\pi}{4}$

MCN331

12. If the area of the triangle formed by the points represented by $z, z + iz$ and iz is 200, then find the value of $|z|$.

MCN332

13. If the complex numbers z_1, z_2 and the z_3 form an equilateral triangle, then prove the following

(a) $\frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$

(b) $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$

(c) $z_1^3 + z_2^2 + z_3^3 = 3 z_1 z_2 z_3$

MCN333

For More Join: @IITJEE_Advanced

14. Consider the complex numbers $z_1 = 10 + 6i$ and $z_2 = 4 + 2i$. If $\arg\left(\frac{z-z_1}{z-z_2}\right)=\frac{\pi}{4}$, find the centre and radius of the locus traced by the complex number z.

MCN334

15. Find all complex numbers common to $|z|=2\sqrt{2}$ and $\arg\left\{\frac{z-(5+i)}{z-(1+i)}\right\}=\frac{\pi}{4}$.

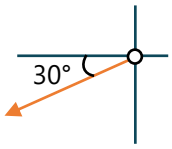
MCN335

Rankers Academy JEE

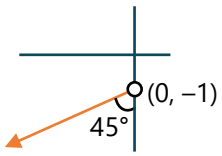
ANSWER KEY

1. A

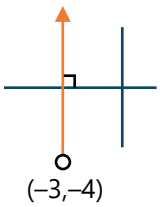
2. (i)



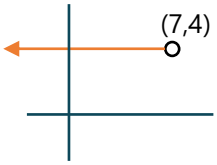
(ii)



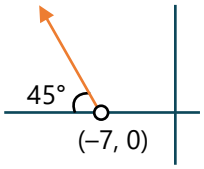
(iii)



(iv)



(v)



3. $2\cos^{-1}\left(\frac{4}{5}\right)$ **5.** 4π **7.** $3e^{i\frac{2\pi}{3}}$ **8.** $1+(2+2\sqrt{2})i$

9. $(\ell < 0, m > 0, n > 0)$ **10.** $\frac{\pi}{2}$ **11.** $\frac{1}{2}(9i-5)$ **12.** 20

14. $5 + 7i$, $\sqrt{26}$ **15.** $z=\left(\frac{3-\sqrt{7}}{2}\right)+i\left(\frac{3+\sqrt{7}}{2}\right)$

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Complex Numbers : RACE-08

Subjective Type Question

1. Solve $z^3 = 8i$ and express its solutions in exponential form.

MCN336

2. Solve $\left(\frac{z}{2}\right)+\left(\frac{z}{2}\right)^2+\left(\frac{z}{2}\right)^3+\left(\frac{z}{2}\right)^4+\left(\frac{z}{2}\right)^5=-1$.

MCN337

3. Solve the equation $(z-2)^{11}=\frac{\sqrt{3}-1+i\left(\sqrt{3}+1\right)}{2\sqrt{2}}$.

MCN338

4. If $\cos\theta + i\sin\theta$ is a root of the equation $x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots\ldots + a_{n-1}x + a_n = 0$ then find the absolute value of $\sum_{r=1}^{n} a_r \cos r\theta$ (where all coefficient are real).

MCN339

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5. Find the sum of the series $1(2-\omega)(2-\omega^2) + 2(3-\omega)(3-\omega^2) \ldots\ldots (n-1)(n-\omega)(n-\omega^2)$ where ω is one of the non real cube roots of unity.

MCN340

6. If ω is a non-real cube root of unity, then prove the following :

(a) $x^3 + y^3 = (x + y)(x + y\omega)(x + y\omega^2)$

(b) $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + y\omega + z\omega^2)(x + y\omega^2 + z\omega)$

(c) $|a+b\omega+c\omega^2| = \sqrt{a^2+b^2+c^2-ab-bc-ca} = |a+b\omega^2+c\omega|$; $a, b, c \in \mathbb{R}$.

MCN341

7. If $z^2 + z + 1 = 0$, where z is complex number, then find the value of

$$\left(z+\frac{1}{z}\right)+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)+\ldots\left(z^6+\frac{1}{z^6}\right)^2.$$

MCN342

8. If l, ω, ω^2 are the cube roots of unity, then prove the following :

(a) $(1 - \omega + \omega^2)^5 + (1 + \omega - \omega^2)^5 = 32$

(b) $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^5) = 1$

(c) $\frac{a+b\omega+c\omega^2}{a\omega+b\omega^2+c} = \omega^2$; $a, b, c \in \mathbb{R}$

MCN343

For More Join: @IITJEE_Advanced

9. If a, b, c, d, e, f, g are the seventh roots of unity and $g \in \mathbb{R}$, then find the value of

(a) $a^2+b^2+c^2+d^2+e^2+f^2+g^2$

(b) $a^3+b^3+c^3+d^3+e^3+f^3$

(c) $a^{-5}+b^{-5}+c^{-5}+d^{-5}+e^{-5}+f^{-5}$

(d) $a^7+b^7+c^7+d^7+e^7+f^7$

(e) $(1+a)(1+b)(1+c)(1+d)(1+e)(1+f)(1+g)$

MCN344

10. If the equation $(z+1)^7 + z^7 = 0$ has roots $z_1, z_2, \ldots z_7$, then find the value of

(a) $\sum_{r=1}^{7} \text{Re}(z_r)$ and (b) $\sum_{r=1}^{7} \text{Im}(z_r)$

MCN345

11. Prove that $\cos\frac{\pi}{7}.\cos\frac{2\pi}{7}.\cos\frac{3\pi}{7}=\frac{1}{8}$ using 7th roots of unity.

MCN346

12. If $1, \alpha_1, \alpha_2 \ldots\ldots, \alpha_{2008}$ are 2009th roots of unity, then find the absolute value of.

$\sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$

MCN347

13. If Z_r ; $r = 1, 2, 3, \ldots, 50$ are the roots of the equation $\sum_{r=0}^{50} (Z)^r = 0$, then find the value of $\left|\sum_{r=1}^{50} \frac{1}{Z_r - 1}\right|$

MCN348

14. If $a_i; i=1,2,3,\ldots,11$ are the eleventh roots of unity, then find $\sum_{i=1}^{10}\left(\sum_{j=i+1}^{11}(a_i a_j)\right)$

MCN349

15. Evaluate $\sum_{p=1}^{32}(3p+2)\left(\sum_{q=1}^{10}\left(\sin\frac{2q\pi}{11} - i\cos\frac{2q\pi}{11}\right)\right)^p$

MCN350

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ANSWER KEY

1. $2\cdot e^{-i\frac{\pi}{2}}, 2\cdot e^{i\frac{\pi}{6}}, 2\cdot e^{i\frac{5\pi}{6}}$ **2.** $2\cdot e^{i\frac{k\pi}{3}}; k=1,2,3,4,5.$ **3.** $2+e^{\frac{i(24n+5)\pi}{132}}; n=0,1,2,\ldots,10$

4. 1 **5.** $\left[\frac{n(n+1)}{2}\right]^2 - n$ **7.** 6

9. (a) 0 (b) –1 (c) –1 (d) 6 (d) 2

10. (a) $-\frac{7}{2}$, (b) zero

12. 2009 **13.** 25 **14.** 0 **15.** $48(1-i)$

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